



**FACULTY OF ENGINEERING, BUILT ENVIRONMENT &
INFORMATION TECHNOLOGY**

**BACHELOR OF INFORMATION TECHNOLOGY (HONS)
BACHELOR OF COMPUTER SCIENCE**

DATABASE FUNDAMENTAL

CONTINUOUS ASSESSMENT

ASSIGNMENT-40%

Submission Deadline: 25/7/2026

CASE STUDY

System Description

SEGi College in Petaling Jaya has a big student's population. There are a number of faculties where each strongly recommended their students to register in one or more clubs and/or societies. The primary goal is to encourage students to participate in clubs and societies to promote good interaction, working effectively with each other and foster creative thinking.

A student can participate in at least one or more clubs. A club must have/elect a president from the club members. An advisor (lecturer) will be appointed to monitor the running of the clubs. One advisor can handle one or more clubs he chose. Each club must run one or more events in a semester.

The student affair department hired you to develop and maintain the database to store all the required data about club memberships and events and will manage the club activities and produce the required reports. Table 1 shows a list of data requirements, which you can use when planning the database. You will need to identify the data to collect, group fields into tables, normalize the data, and relate the tables. Table 1 is the data requirements given to you. You are the only person responsible for the database and there are many responsibilities for you to provide a database that works well for the department.

Student Affair department decides that the best way to get started planning the database is to prepare a list of all the data items that are needed to support the department.

Table 1 Data Description

1. Students: to store information about students who register to clubs. Students can sign up for more than one clubs.	
StudentID number	Student are 2 letters and 4 digits ex; hh1234
Name	Full Name
Phone	Mobile phone number.
Faculty	The student's faculty.
Faculty Approval Form	Students must get approval from faculty
Scholarship	We keep the record if the student received a scholarship (Yes/No)
2. Clubs	
Name	Clubs including IT Club, Dance Club, Music Club, Accounting Club (any clubs you wish to add) There are several clubs available, provided in the link below: https://colleges.segi.edu.my/kotadamansara/student-clubs-societies/
Advisor	Each club has an assigned advisor. We need to know the advisor's name, office room number (eg: R2.1), office phone number. An advisor can advise one or more clubs.
President	From student members
Date register	Date the students become member
Notes	A place to record notes about each club.
3. Venue: Different rooms, halls are available for events.	
ID number	Rooms and halls are assigned ID numbers.
Type	Type of room, hall etc.
Staff in charge	Each venue has an assigned person in charge (PIC), who manages and resolves problems with the venue. We will need to know the person's name, phone number, and office number.
4. Event: For each club event, we need to know the club event number, club president, activity name, semester number and the venue needed for the activity.	

TASK 1: Database Design – 40 marks

One of the team members has done preliminary requirements analysis and prepared some documentation for you to further analyse and design. Refer to System Description and Table 1 Data Description for you to complete the required tasks.

1. Summarize the **data description** above to show relationships between entities (multiplicities)

Entity Name	Multiplicity/ Cardinality	Relationship name	Entity Name	Description
Example: Customer	1..1	places	Order	A customer who places order

(10 marks)

2. You also need to create a **Data Dictionary** based on Question 1 to show the final structure of the database. The table should include the following columns:

Entity Name	Attribute	Data Type & Length	Constraints (PK, FK)	Null/Not Null
Customer	C_name C_ID	Varchar2(50) integer	Primary key	Not null Not null

(10 marks)

3. From the data dictionary obtained above, you need to draw a complete **Entity Relationship Diagram** with clear entities, attributes, relationships, primary key, foreign key, junction table and cardinalities. You can use any available online CASE tools (draw.io, lucidchart).

(20 marks)

TASK 2: Data Definition Language (DDL) - 30 marks

1. From the entities identified during data design, you are to create a database using SQL. Use Oracle /Oracle Developer/SQL Live Oracle to create a database with all the necessary tables and constraints (PK, FK, not null, unique, check etc). Include the primary keys and foreign keys apparent in the tables specified above.
(15 marks)
2. Now that you have all the tables set up, populate the tables with relevant and logical/meaningful data. For each table, enter at least 10 rows where applicable.
(15 marks)

TASK 3: Data Manipulation Language (DML)/Queries – 30 marks

In this section, you will need to develop a number of queries for use by the department.

1. The staff often needs to contact students who have enrolled in any clubs. Create a phone list the staff can use. The list should be ordered so that it is easy to find a student by name.
(5 marks)
2. Because the advisor (lecturer) serves as an advisor in addition to their other responsibilities, the lecturer needs to schedule their time carefully. In particular, they need to monitor people who advise more than one club. You are asked to list advisors who are assigned to more than one club and to identify the clubs to which each advisor is assigned.
(5 marks)
3. Before students can participate in a club, they must provide a faculty approval form. One staff member is assigned to calling students who are missing a required form. You need to create a list, including all phone numbers, of students who are missing the required forms. The list should identify the missing form.
(5 marks)

4. The department needs a list of advisors assigned to the clubs. An advisor often wants a quick reference to their clubs and events, and the date of the events.

(5 marks)

5. The staff also needs to know how many clubs each advisor is assigned to advise for each semester. They ask you to provide this information in the form of a spreadsheet, with the advisors' names appearing for the row labels and the semesters appearing as the column headings. The cell contents should indicate the number of events each advisor is assigned to in each semester.

(5 marks)

6. All students who have signed up to attend an event should be assigned to a club. Provide a list of clubs and the students assigned to those clubs.

(5 marks)

TASK 4: Demonstration of the Assignment

- Please prepare to turn on your Oracle database and your **assignment report** during the demo.
- **Date: Week 11(4th Live Session)**

Assessment Marking Criteria

GRADE	Data Modeling	Definition & Insertion Using SQL & Data Manipulation Using SQL (Implementation)
A – 75% -100%	A deep understanding of process, logic and data modelling is evident from the coherence, correctness and complexity of the models. The detailed description of the relationship between the ERD particularly indicates tacit knowledge of data modelling and high-level of understanding of model integration. At the very best, the student shows exceptionally high levels of creative and analytical problem-solving skills that makes use of information that is beyond that given in the module.	A convincingly proficient use of Oracle SQL is demonstrated with <i>all</i> 6 queries being accurately addressed. It is clear the group understand the underlying processing of a SELECT statement.
B – 60% - 74%	All taught techniques have been competently applied to model the requirements. Competent use of drawing tools and notation. However, the techniques use and information given typically remains within the confines of the taught material.	A persuasive use of Oracle SQL to address <i>all</i> of required queries. All attributes are implemented as defined and <i>most</i> integrity rules have been accurately implemented (with some initial values defined.) Foreign key constraints have been added correctly. A strong understanding of the various issues pertaining to referential integrity is also evident in places – as is how to enforce it using Oracle SQL (using keywords such as UPDATE, DELETE, CASCADE). The group is typically able to obtain meaningful rows and columns using attribute renaming, arithmetic functions, compound logical expressions, joins, simple correlated subqueries and compound subqueries. All explanations and results are presented in a well- structured manner and coherently described in detail.
C+ – 55% - 59%	The process, data and logic models are produced using appropriate drawing tools and mainly use an agreed notation with some accuracy Correct data types have typically been assigned to the attributes and most primary and foreign key constraints have been correctly implemented. Some gaps in Oracle/SQL knowledge may typically be apparent for implementing compound keys and enforcing data integrity during data definition.	Good use of Oracle SQL to address at least <i>three</i> of the required queries. Although one or two queries presented may be incomplete, the group is typically able to obtain meaningful rows and columns using attribute renaming, arithmetic functions, compound logical expressions and/or joins.
C – 50% - 54%	Typically, one type of logical system model is produced more accurately than the others. Organisation and coherence of the models are limited and show little integration. Further study is clearly required to aid ability to construct system models	An acceptable attempt has been made to use SQL to address the first two queries required. Although some of the queries presented are incomplete, it adequately makes use of the SELECT statement to obtain meaningful rows and columns.

	that are balanced, fully normalised and integrated.	An acceptable attempt has been made to implement the data model using SQL. Some of the table definitions have been correctly implemented and some of the relationships at least partially established.
F – 0 – 49%	At best, some useful/relevant information may be present but much more academic study is clearly required for it to be adequate.	There is insufficient evidence that Oracle SQL has been sufficiently understood to enable the group to adequately perform the majority of the required queries. Most of the queries do not work at all or have not been attempted. Insufficient evidence to show that the appropriate ORACLE SQL commands for creating the required tables, attributes and constraints have been adequately understood and applied. Although, a number of CREATE TABLE/INSERT INTO statements may exist, it typically addresses only a small part of the table specification.

Marking Rubric